DC Characterization of an NMOS Transistor in 180 nm CMOS Technology

Toney
VLSI Design Project – Cadence Virtuoso

July 1, 2026

Contents

Abstract

This report presents the DC characterization of an NMOS transistor implemented in a 180 nm CMOS process using Cadence Virtuoso Schematic Editor and Spectre simulation. The device under test has a channel width $W = 2\,\mu$m and channel length $L = 180\,$nm. A DC sweep of the drain-to-source voltage V_{DS} from 0 to 2 V was performed for five different gate-to-source bias voltages $V_{GS} \in \{0, 0.5, 1.0, 1.5, 2.0\}$ V, producing the classic family of I_D–V_{DS} output characteristic curves. The resulting curves are analyzed to illustrate the triode and saturation regions of MOSFET operation and to extract qualitative trends consistent with square-law MOSFET behavior.

1 Objective

The goal of this exercise is to characterize the output (I_D vs. V_{DS}) behavior of a single NMOS transistor in a 180 nm technology node, and to observe how the drain current changes as a function of both the drain-source voltage and the gate-source overdrive voltage. This characterization is a fundamental first step in analog/mixed-signal IC design, as it is used to identify the threshold voltage, the triode-to-saturation transition, and the transconductance behavior of the device, which subsequently inform sizing decisions for amplifiers, current mirrors, and switches.

2 Simulation Setup

2.1 Schematic

The test bench, shown in Figure 1, consists of a single NMOS transistor (`nmos1`) with the following sizing:

- Width: $W = 2\,\mu\text{m}$
- Length: $L = 180\,\text{nm}$
- Multiplier: $m = 1$

Two independent DC voltage sources bias the device:

- **V1**: connected between the drain node and ground, with $V_{DC} = V_{DS}$, used as the swept sweep variable for the DC analysis.
- **Vgs**: connected between the gate node and ground, with $V_{DC} = V_{GS}$, used as the parametric (outer-loop) sweep variable.

The source and bulk/body terminal of the NMOS are tied to ground, and the drain is connected to the top of source `V1` (i.e., V_{DS} is applied directly across drain-source).

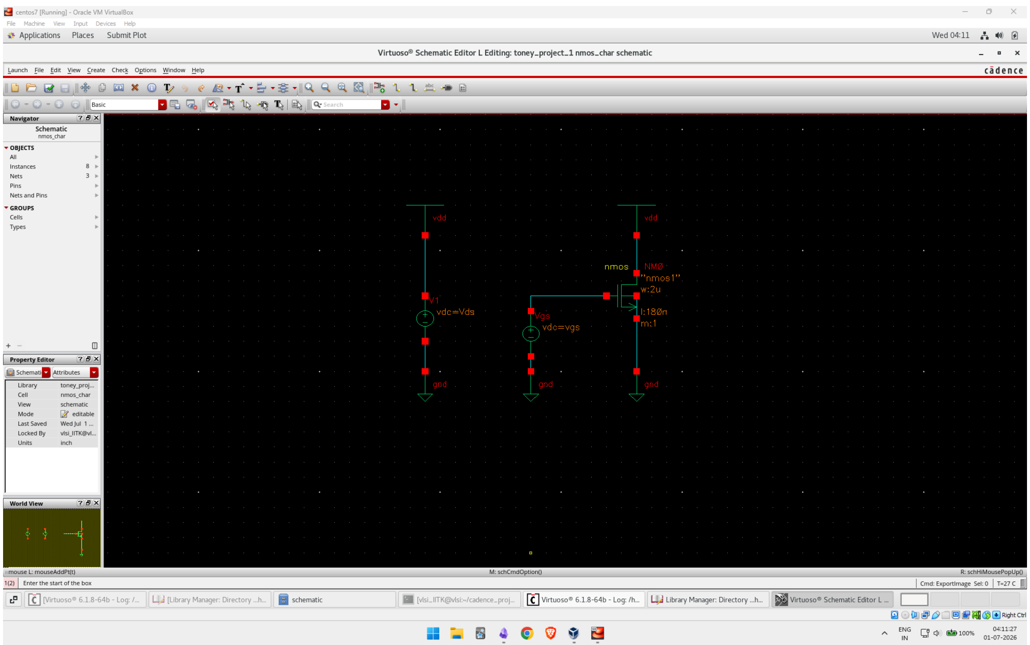

Figure 1: NMOS characterization test bench in Cadence Virtuoso Schematic Editor. The transistor is sized W=2μm, L=180nm, with independent DC sources biasing V_{DS} and V_{GS}.

2.2 Default Operating Point and Sweep Definitions

Before running the parametric sweep, the default (nominal) bias point of the circuit was set as follows:

- $V_{GS,\text{default}} = 0.9\,\text{V}$
- $V_{DS,\text{default}} = 0.9\,\text{V}$

The DC simulation was then configured with two nested sweeps:

Table 1: DC and Parametric Sweep Configuration

Variable	**Sweep Type**	**Range**	**Step**
V_{DS}	Primary DC sweep	$0\,\text{V} \to 2\,\text{V}$	Linear (fine)
V_{GS}	Parametric (outer) sweep	$0\,\text{V} \to 2\,\text{V}$	$0.5\,\text{V}$

This configuration produces five output curves, corresponding to $V_{GS} = 0,\ 0.5,\ 1.0,\ 1.5,$ and $2.0\,\text{V}$, each plotted as I_D versus V_{DS} over the full 0–2 V range.

3 Results

Figure 2 shows the simulated I_D–V_{DS} output characteristics for the five values of V_{GS}.

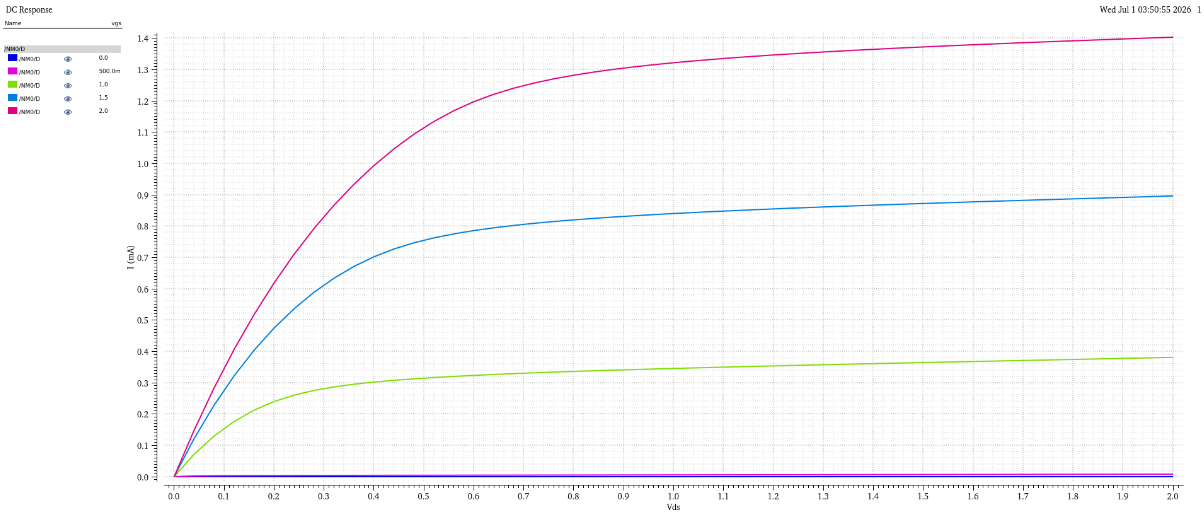

Figure 2: Simulated I_D vs. V_{DS} characteristics of the NMOS device for $V_{GS} = 0,\ 0.5,\ 1.0,\ 1.5,\ 2.0\,\text{V}$.

3.1 Observed Trends

Table 2: Approximate drain current at $V_{DS} = 2\,\text{V}$ for each V_{GS} value (read from simulation results)

V_{GS} (V)	Region of Operation	I_D at $V_{DS} = 2\,\text{V}$
0.0	Cut-off	$\approx 0\,\mu\text{A}$
0.5	Near/sub-threshold	$\approx 10\,\mu\text{A}$
1.0	Saturation	$\approx 0.38\,\text{mA}$
1.5	Saturation	$\approx 0.90\,\text{mA}$
2.0	Saturation	$\approx 1.41\,\text{mA}$

Several characteristic MOSFET behaviors are visible in Figure 2:

1. **Cut-off region:** For $V_{GS} = 0\,\text{V}$ (and to a large extent $V_{GS} = 0.5\,\text{V}$), the drain current remains essentially zero across the entire V_{DS} range, indicating that the gate bias is below (or only marginally above) the threshold voltage V_{TH} of the device. This places an estimate of V_{TH} somewhere between 0.5 V and 1.0 V, consistent with a typical 180 nm NMOS threshold voltage of roughly 0.4–0.5 V once the small but nonzero current at $V_{GS} = 0.5\,\text{V}$ is taken into account.

2. **Triode (linear) region:** For each curve with $V_{GS} > V_{TH}$ (i.e., $V_{GS} = 1.0, 1.5, 2.0\,\text{V}$), the drain current rises sharply and nearly linearly for small V_{DS} (roughly $V_{DS} < V_{GS} - V_{TH}$), where the transistor behaves like a voltage-controlled resistor.

3. **Saturation region:** Beyond the triode region, each curve bends and flattens out, transitioning into saturation once $V_{DS} \geq V_{GS} - V_{TH}$. The drain current becomes only weakly dependent on V_{DS}, with a small positive slope due to channel-length modulation (finite output resistance r_o).

4. **Square-law dependence on V_{GS}:** The saturation current increases super-linearly with V_{GS}. Going from $V_{GS} = 1.0\,\text{V}$ to 1.5 V (+0.5 V) increases I_D by roughly 0.52 mA, while going from 1.5 V to 2.0 V increases I_D by roughly 0.51 mA. The ratio of currents is broadly consistent with the square-law MOSFET model, $I_D \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2(1 + \lambda V_{DS})$, since the overdrive voltage $(V_{GS} - V_{TH})$ roughly doubles between the $V_{GS} = 1.0\,\text{V}$ and $V_{GS} = 2.0\,\text{V}$ curves while I_D increases by roughly $3.7\times$.

4 Discussion

The family of curves obtained matches the expected qualitative behavior of a long-channel MOSFET output characteristic. The clear separation between cut-off, triode, and saturation regions confirms correct device connectivity (source/bulk grounded, gate and drain independently biased) and correct sweep configuration (inner sweep on V_{DS}, outer/parametric sweep on V_{GS}).

The mild upward slope of the curves in saturation (most visible for $V_{GS} = 2.0\,\text{V}$, where I_D increases from about 1.30 mA at $V_{DS} = 1\,\text{V}$ to about 1.41 mA at $V_{DS} = 2\,\text{V}$) is attributable to channel-length modulation, and would be expected to be more pronounced for shorter channel lengths – consistent with the relatively short $L = 180\,\text{nm}$ used in this device.

5 Conclusion

The I_D–V_{DS} characterization of the 180 nm NMOS transistor was successfully performed in Cadence Virtuoso using a DC sweep of V_{DS} from 0 to 2 V, parametrized over V_{GS} from 0 to 2 V in 0.5 V steps. The resulting curves clearly show the cut-off, triode, and saturation regions of operation, an approximate threshold voltage between 0.5 and 1.0 V, and saturation currents ranging from near zero (at $V_{GS} = 0\,\mathrm{V}$) up to approximately 1.41 mA (at $V_{GS} = 2.0\,\mathrm{V}$, $V_{DS} = 2.0\,\mathrm{V}$). These results validate the basic functionality of the device model and test bench, and provide a baseline characterization that can be used for subsequent analog circuit design (e.g., biasing, sizing for target g_m or I_D).